Scalability&FuturePlan

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TeamID	SWUID2026221635
ProjectName	A Comprehensive Measure of Well-Being (HDI Prediction & Analysis)
MaximumMarks	1Mark

Scalability&FuturePlan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High/Medium/Low)
1	Static Model File (.pkl)	The Linear Regression model cannot automatically learn from new country data unless a developer manually retrains and replaces the pickle file.	High
2	Local Database (Mongo DB/CSV)	Data storage is confined to a local environment, preventing multi-user synchronization and global data updates.	Medium
3	Single-Threaded Server	Flask's default development server runs on a single thread, causing page load drops if multiple users drag input sliders at the same time.	High

ScalabilityPlan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Runs locally on a single-threaded Flask server handling very few concurrent users.	Deploy the application using an asynchronous WSGI server like **Gunicorn** or **uWSGI** hosted on cloud platforms like Render or AWS to handle hundreds of active sessions.
2	Data Storage	Uses structured text format (CSV files) or standalone SQLite database.	Migrate the backend database storage to an enterprise-grade cloud database like **PostgreSQL** or Amazon RDS to handle large global datasets efficiently.

3	Performance	Matplotlib/Seaborn charts are rendered sequentially on the server side, which uses processing memory.	Offload chart visualizations to frontend JavaScript frameworks (like **Chart.js** or D3.js) by passing raw JSON scores from the Flask API, boosting rendering speed.
4	Security	Standard localized session variables and basic Werkzeug security checking.	Implement robust token-based authentication using **Flask-JWT-Extended** and secure the entire data stream with HTTPS/SSL layers.

FutureRoadmap:

Phase	Planned Feature / Enhancement	TargetTimeline	Expected Impact
Phase 2	Integrate the **World Bank Open Data API** to automatically pull the latest annual global education, health, and GNI metrics without manual file uploads.	Next 3 Months	Eliminates operational overhead and ensures the predictive environment is always running on up-to-date statistics.
Phase 3	Upgrade the core engine from simple Linear Regression to **Polynomial Regression** or **Ensemble Methods (Random Forest Regressor)** to capture complex non-linear economic factors.	Next 6 Months	Increases the continuous HDI score calculation accuracy across developing and highly developed countries alike.
Phase 4	Build a "What-If" multi-country simulation dashboard allowing users to compare side-by-side scenarios for up to 3 countries.	Next 9 Months	Transforms the tool into a powerful multi-tenant collaborative platform tailored for international policy experts and researchers.